exert on each other. The situation in general relativity is much the same, except that Einstein's equations show that Newton's focus on mass was too limited. According to general relativity, it is not just the mass (and the separation) of objects that contributes to the strength of the gravitational field. *Energy* and *pressure* also contribute. This is important, so let's spend a moment to see what it means.

Imagine that it's the twenty-fifth century and you're being held in the Hall of Wits, the newest Department of Corrections experiment employing a meritocratic approach to disciplining white-collar felons. The convicts are each given a puzzle, and they can regain their freedom only by solving it. The guy in the cell next to you has to figure out why *Gilligan's Island* reruns made a surprise comeback in the twenty-second century and have been the most popular show ever since, so he's likely to be calling the Hall home for quite some time. Your puzzle is simpler. You are given two identical solid gold cubes—they are the same size and each is made from precisely the same quantity of gold. Your challenge is to find a way to make the cubes register different weights when gently resting on a fixed, exquisitely accurate scale, subject to one stipulation: you're not allowed to change the amount of matter in either cube, so there's to be no chipping, scraping, soldering, shaving, etc. If you posed this puzzle to Newton, he'd immediately declare it to have no solution. According to Newton's laws, identical quantities of gold translate into identical masses. And since each cube will rest on the same, fixed scale, earth's gravitational pull on them will be identical. Newton would conclude that the two cubes must register an identical weight, no ifs, ands, or buts.

With your twenty-fifth-century high school knowledge of general relativity, though, you see a way out. General relativity shows that the strength of the gravitational attraction between two objects does not just depend on their masses[5] (and their separation), but also on any and all additional contributions to each object's total *energy.* And so far we have said nothing about the temperature of the golden cubes. Temperature is a measure of how quickly, on average, the atoms of gold that make up each cube are moving to and fro—it's a measure of how energetic the atoms are (it reflects their *kinetic* energy). Thus, you realize that if you heat up one cube, its atoms will be more energetic, so it will weigh a bit more than the cooler cube. This is a fact Newton was unaware of (an